

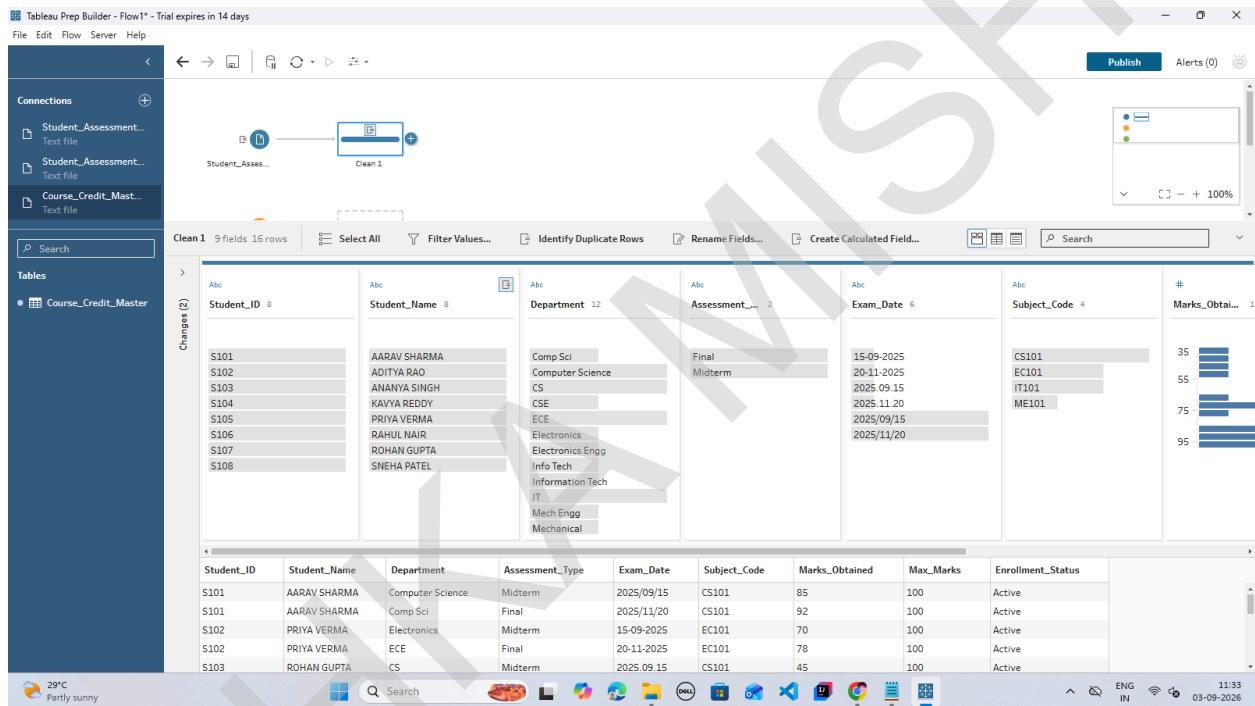
SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 10

AIM: Lesson 10: Data Preparation and Transformation using Tableau Prep

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

A: Connect and Load Data into Tableau Prep



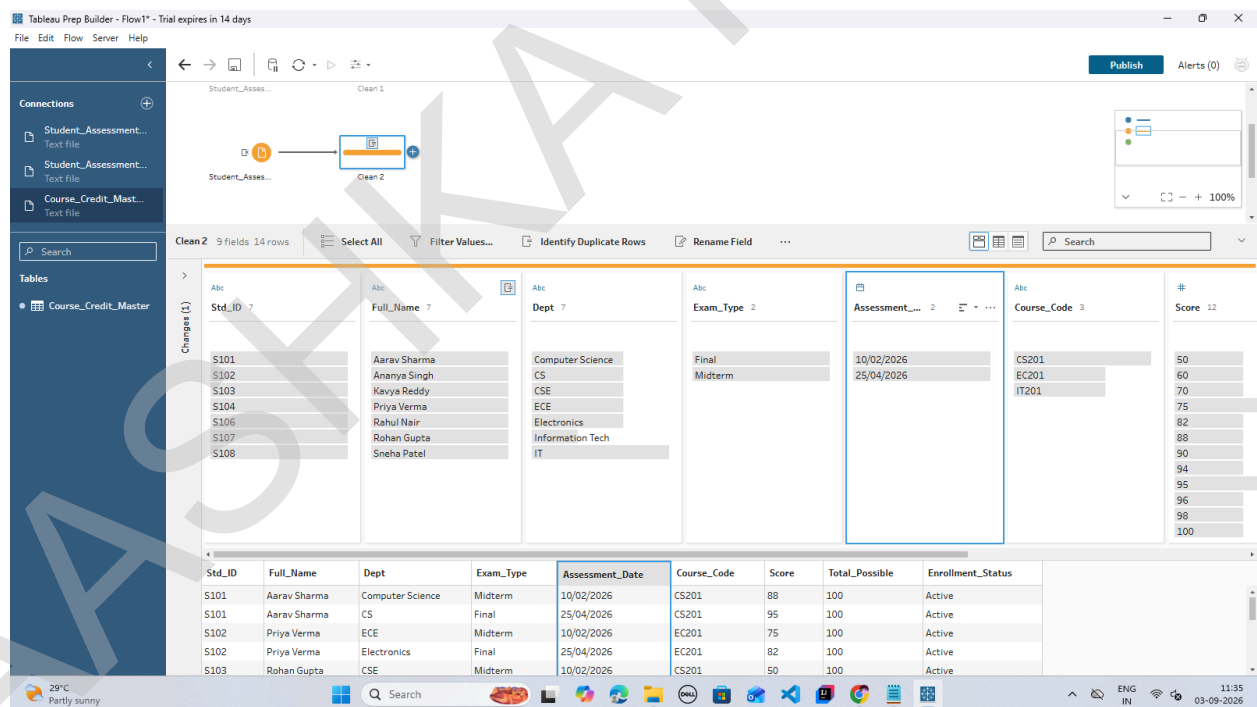
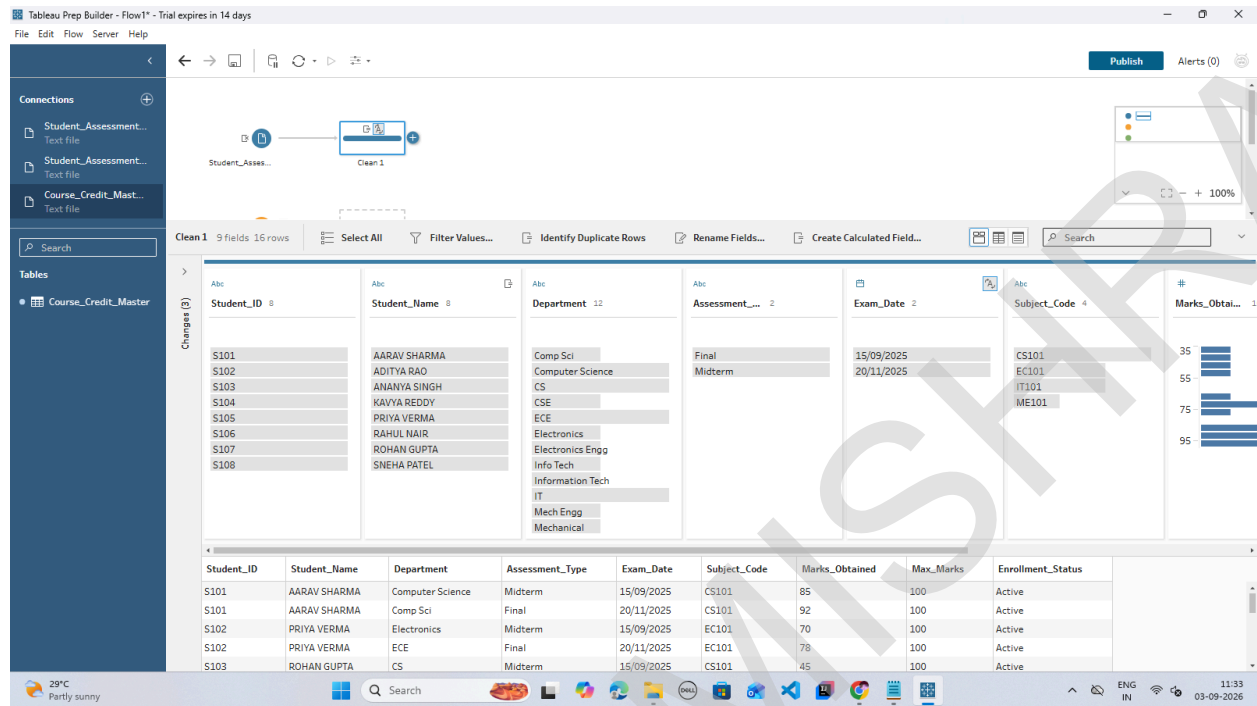
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ROLL NO: T095

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B: Clean and Profile Data

Step 1: Inspect the Profile Pane



A: Connect and Load Data into Tableau Prep

B: Clean and Profile Data

C: Group and Standardize Values

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Step 1: Inspect the Department Field in Term 1 (Clean 1)

Step 2: Group Values Using Pronunciation / Spelling or Manual Selection

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Tables

- Course_Credit_Master

Changes (4)

Student_ID 8

Student_Name 8

Department 6

Assessment_Type 2

Exam_Date 3

Group Values by Manual Selection

Done

Mechanical 2 members

- ☒ Mech Engg
- ☒ Mechanical

Computer Science

Electronics

Info Tech

Information Tech

IT

Final

Midterm

15/09/2025

20/11/2025

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S105	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
S105	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped

29°C Partly sunny

Search

ENG IN

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Step 3: Repeat Grouping on Term 2 (Clean 2)

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Tables

- Course_Credit_Master

Changes (2)

Std_ID 7

Full_Name 7

Dept 3

Exam_Type 2

Assessment_Type 2

Group Values by Manual Selection

Done

Information Tech 2 members

- ☒ Information Tech
- ☒ IT

CS

Electronics

Information Tech

IT

Final

Midterm

10/02/2026

25/04/2026

Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S104	Sneha Patel	Information Tech	Final	25/04/2026	IT201	98	100	Active
S108	Kavya Reddy	Information Tech	Midterm	10/02/2026	IT201	95	100	Active
S108	Kavya Reddy	Information Tech	Final	25/04/2026	IT201	100	100	Active
S104	Sneha Patel	Information Tech	Midterm	10/02/2026	IT201	94	100	Active

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D: Merge Mismatched Fields

Step 1: Create the Union Step on the Canvas

Step 2: Inspect Mismatched Fields in the Union Pane

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists 'Student_Assessment...' and 'Course_Credit_Mast...'. The 'Tables' pane shows 'Course_Credit_Master'. The main canvas displays a flow: 'Student_Asses...' -> 'Clean 1' -> 'Union 1'. Below the canvas, the 'Settings' pane for 'Union 1' shows 'Inputs' as 'Clean 1' and 'Clean 2'. The 'Resulting Fields' section indicates '16 Mismatching fields from 18 resulting fields'. The 'Mismatched Fields' list includes 'Full_Name', 'Student_Name', 'Student_ID', 'Department', 'Assessment_Type', 'Exam_Date', and 'Subject_Code'. The 'Union Results' pane shows a table with columns: 'Student_ID', 'Student_Name', 'Department', 'Assessment_Type', and 'Exam_Date'. The table contains data for 10 rows, with some fields being null.

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC
ABC	ABC	ABC	ABC	ABC

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists 'Student_Assessment...' and 'Course_Credit_Mast...'. The 'Tables' pane shows 'Course_Credit_Master'. The main canvas displays a flow: 'Student_Asses...' -> 'Clean 1' -> 'Union 1'. Below the canvas, the 'Settings' pane for 'Union 1' shows 'Inputs' as 'Clean 1' and 'Clean 2'. The 'Resulting Fields' section indicates '0 Mismatching fields from 10 resulting fields'. The 'Union Results' pane shows a table with columns: 'Enrollment_Status', 'Dept', 'Exam_Type', 'Assessment_Date', and 'Course_Code'. The table contains data for 10 rows, with some fields being null.

Enrollment_Status	Dept	Exam_Type	Assessment_Date	Course_Code
Active	Computer Science	Final	15/09/2025	CS101
Dropped	CS	Midterm	20/11/2025	CS201
	Electronics		10/02/2026	EC101
	Info Tech		25/04/2026	EC201
	Information Tech			IT101
	Mechanical			ME101

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Step 4: Add a Clean Step Post-Union

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Tables

- Course_Credit_Master

Flow Canvas:

- Student_Asses... (Clean 1) → Union 1 → Clean 5

Data Preview (Clean 5):

Full_Name	Table Names	Enrollment_Status	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Std_ID
Aarav Sharma	Student_Assessments_Term1	Active	Computer Science	Final	15/09/2025	CS201	88	100	S101
AARAV SHARMA	Student_Assessments_Term2	Dropped	CS	Midterm	20/11/2025	CS201	95	100	S101
ADITYA RAO			Electronics		10/02/2026	EC201	75	100	S102
Ananya Singh			Info Tech						
ANANYA SINGH			Information Tech						
Kavya Reddy			Mechanical						
KAVYA REDDY									
Priya Verma									
PRIYA VERMA									
Rahul Nair									
RAHUL NAIR									
Rohan Gupta									

E: Perform Aggregations in Prep

Step 1: Add a Term Identifier Field in Clean 3

Step 2: Add an Aggregate Step to the Flow Canvas

Step 3: Configure Grouped Fields (Dimensions)

Tableau Prep Builder - Flow1* - Trial expires in 14 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text File
- Student_Assessment... Text File
- Course_Credit_Mast... Text File

Tables

- Course_Credit_Master

Flow Canvas:

- Student_Asses... (Clean 1) → Union 1 → Clean 5 → Aggregate 1

Settings (Aggregate 1):

Additional Fields

Drag fields to aggregate or group them.

Buttons: Add All, Remove All

Fields:

- GROUP: Assessment_Date
- GROUP: Course_Code
- GROUP: Enrollment_Status
- GROUP: Exam_Type
- SUM: Score
- GROUP: Table Names
- SUM: Total_Possible
- SUM: Number of Rows (Aggregated)

Grouped Fields

GROUP	GROUP
Dept	Std_ID
Computer Science	S101
CS	S102
Electronics	S103
Info Tech	S104
Information Tech	S105
Mechanical	S106
	S107
	S108

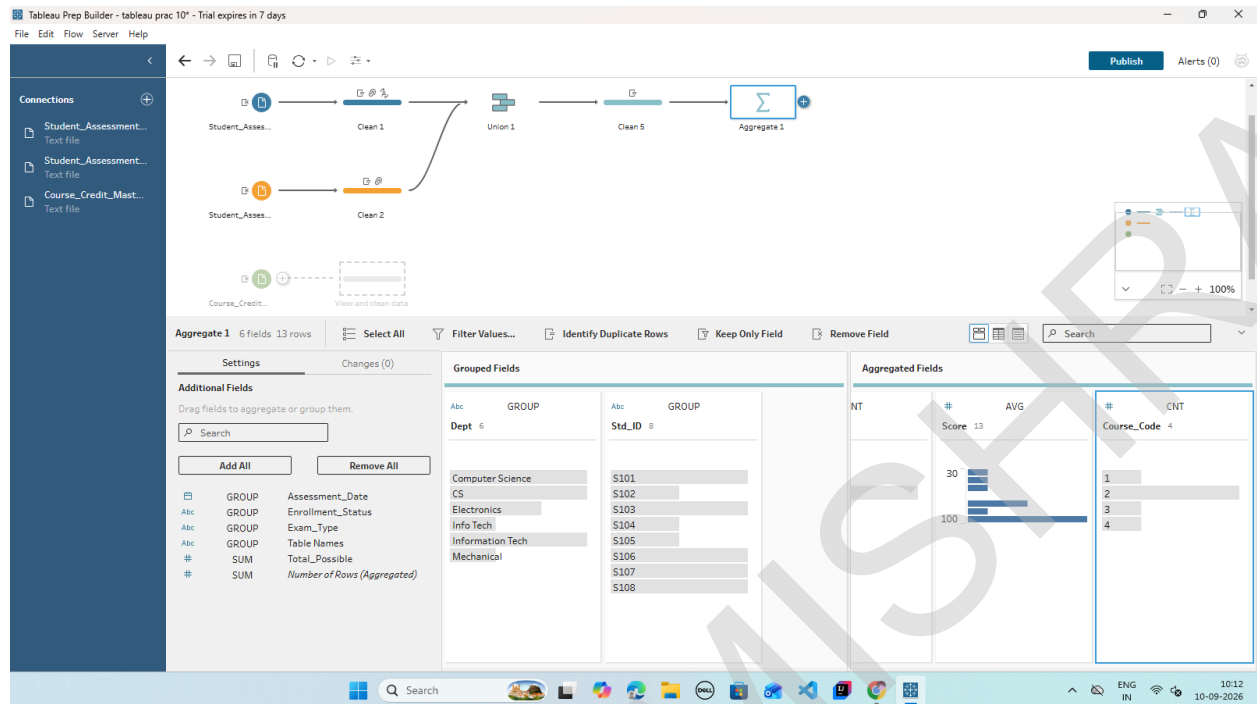
Aggregated Fields

CNT	CNT
Term	Full_Name
1	1
2	2
3	3
4	4

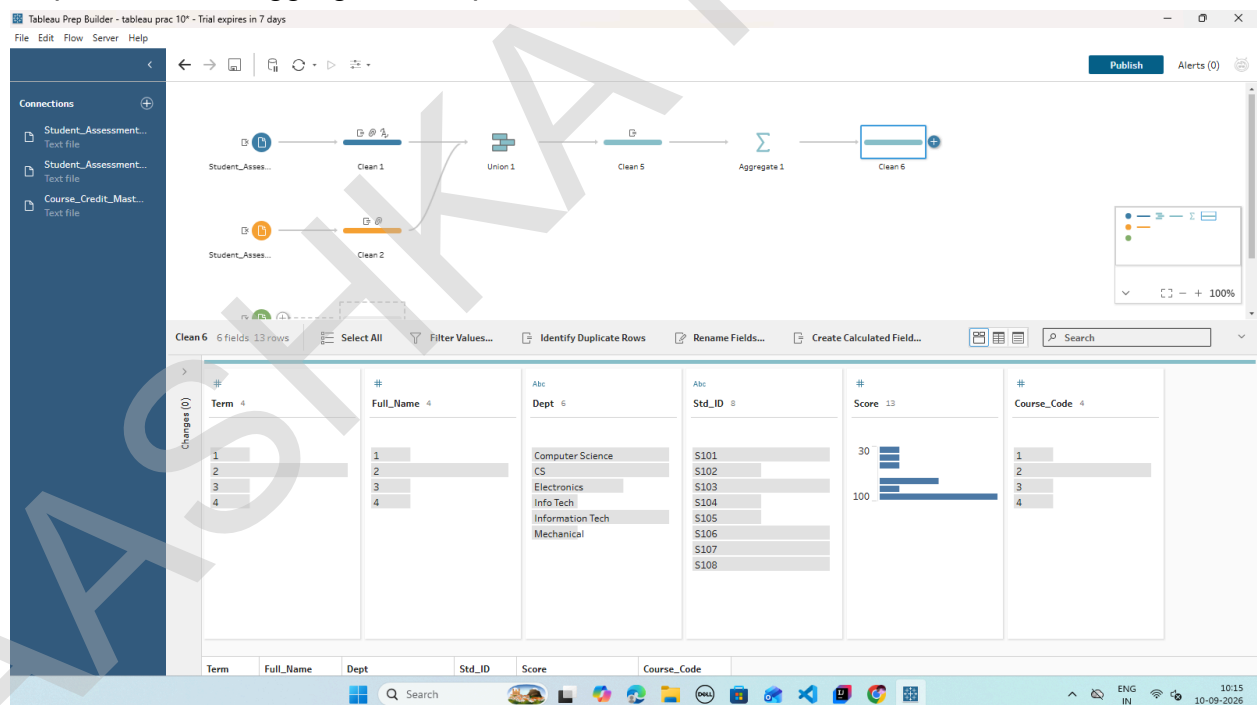
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Step 4: Configure Aggregated Fields (Measures & Metrics)



Step 5: Rename Aggregated Output Fields



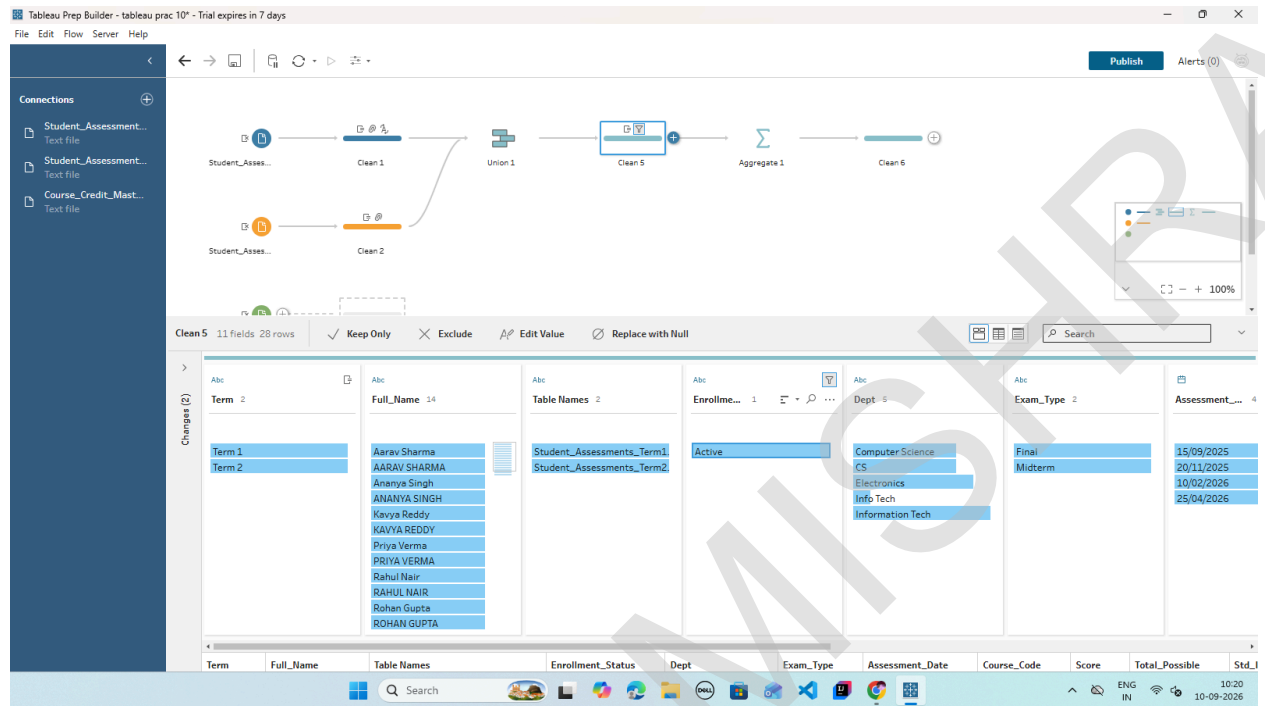
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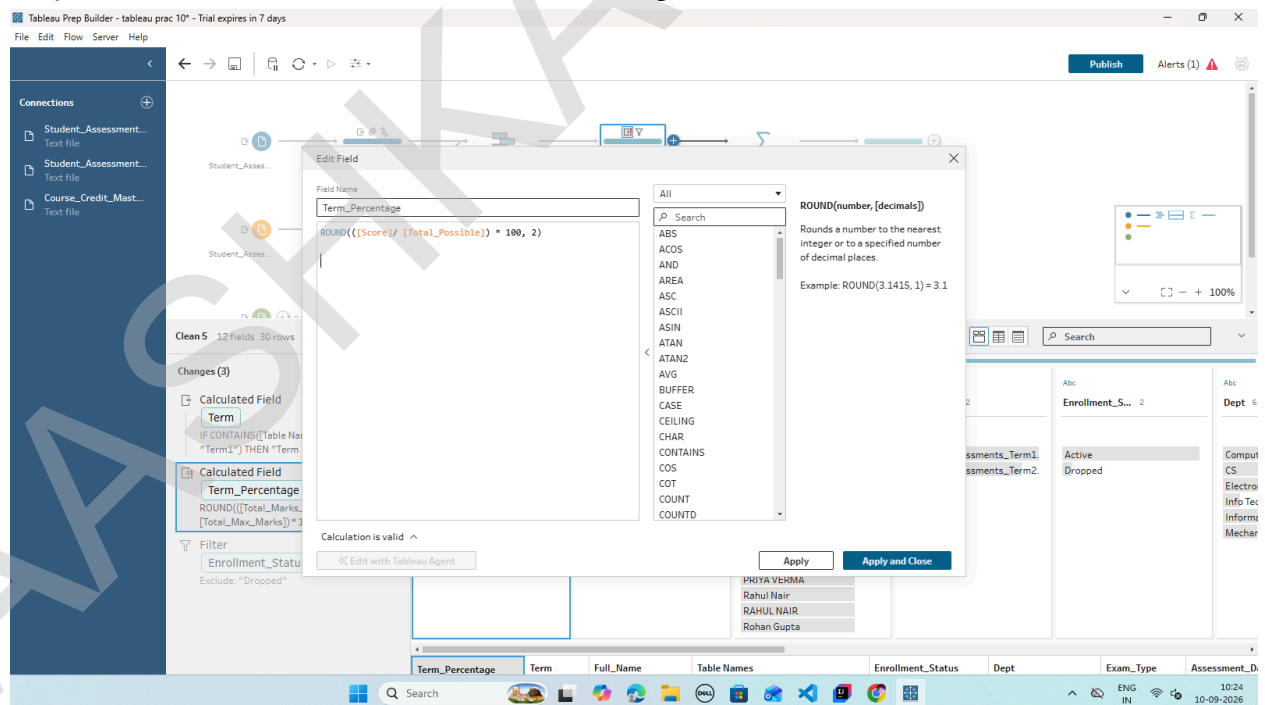
F: Apply Filters and Transformations

Step 1: Open the Transformation Step (Clean 4)

Step 1: Open the Transformation Step (Clean 4)

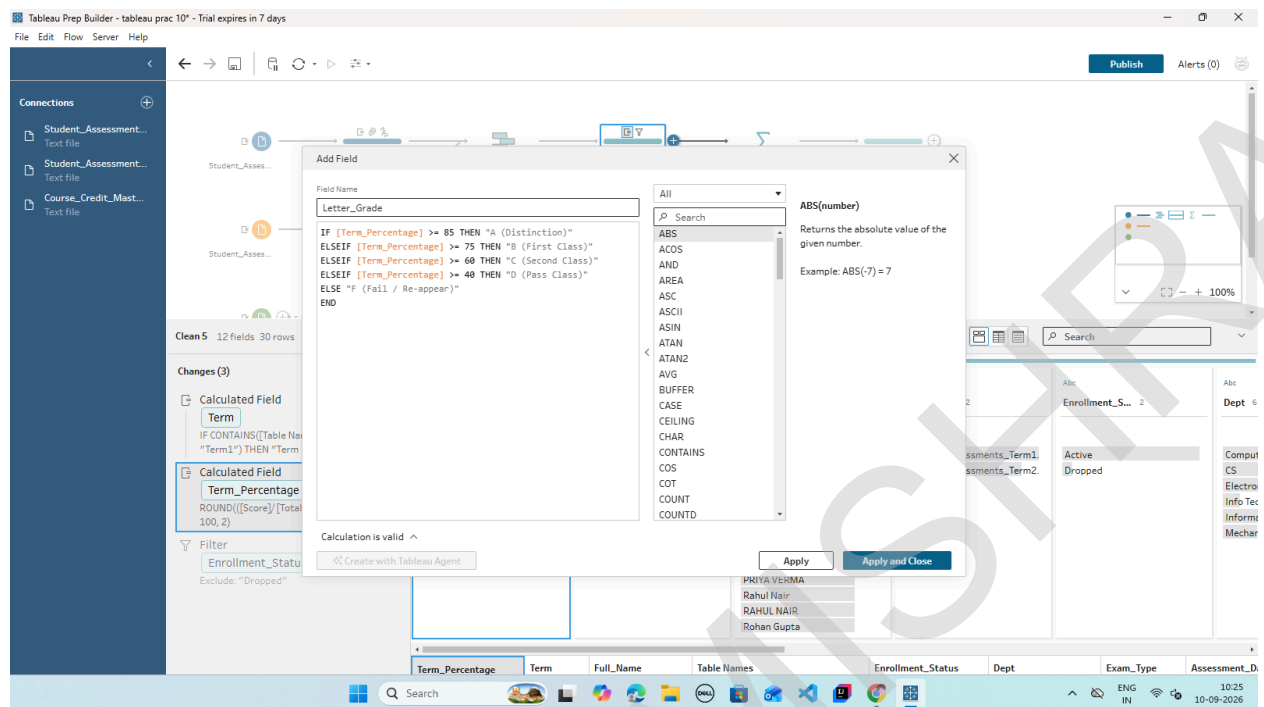


Step 3: Create the Normalized Term Percentage Metric

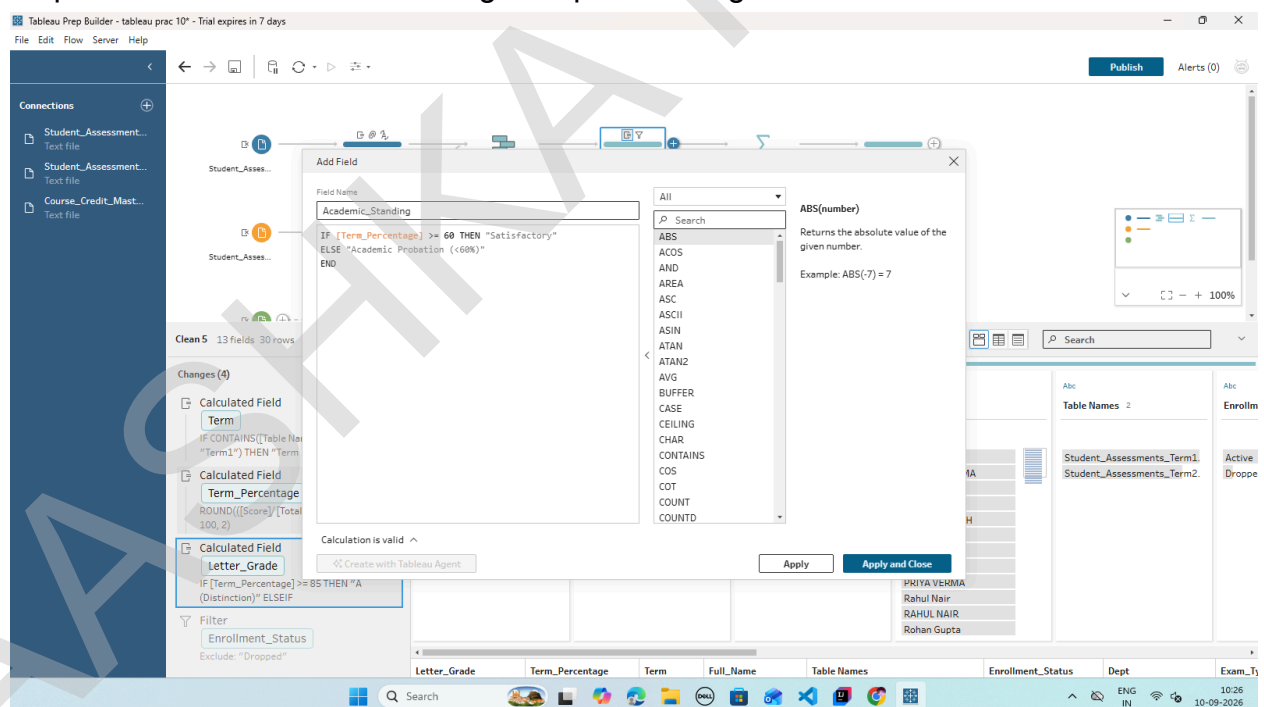


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Step 4: Create the Categorical Letter Grade Transformation



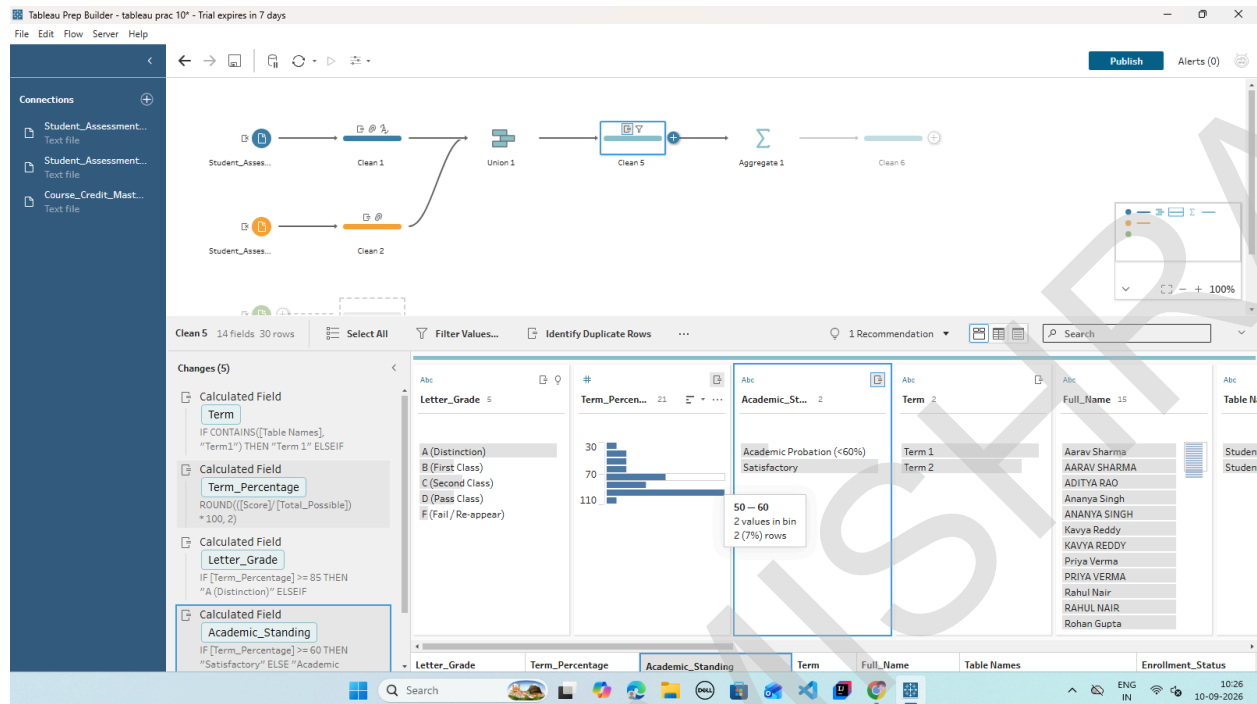
Step 5: Create Academic Standing Compliance Flag



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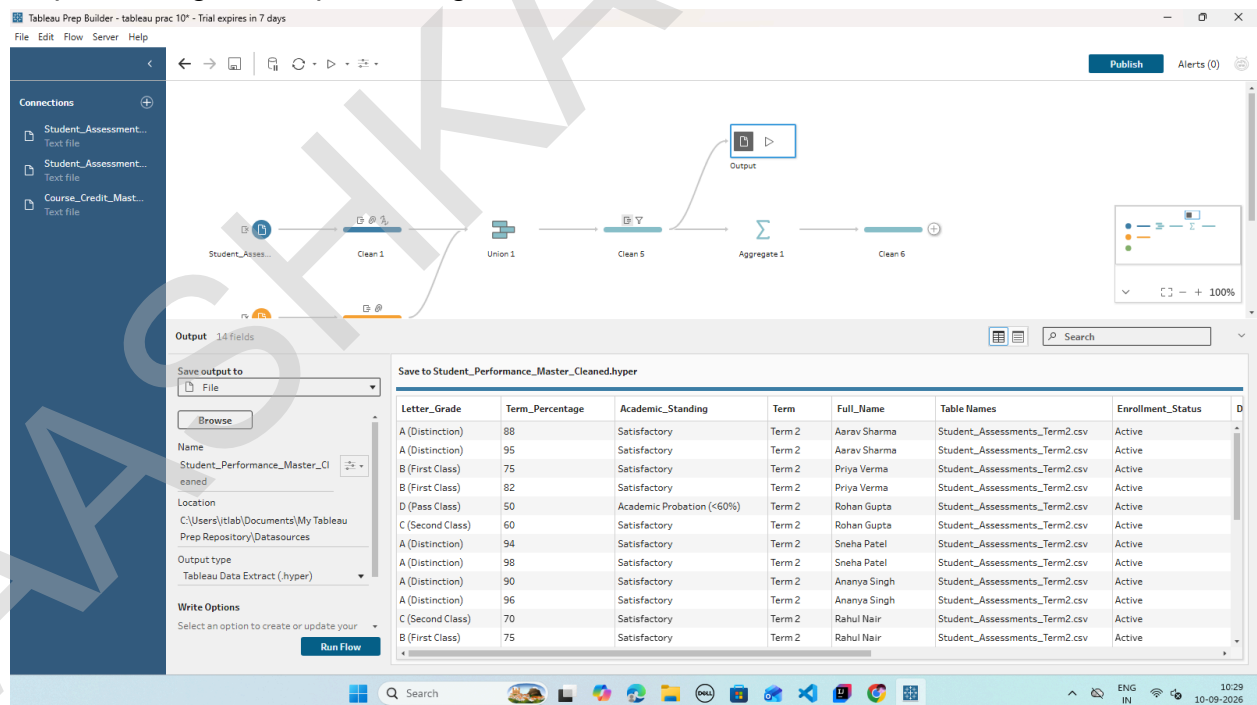
Step 6: Verify Transformed Profile Distributions



G: Create and Automate Data Preparation Flows

Step 1: Add an Output Step to the End of the Pipeline

Step 2: Configure Output Settings in the Bottom Pane

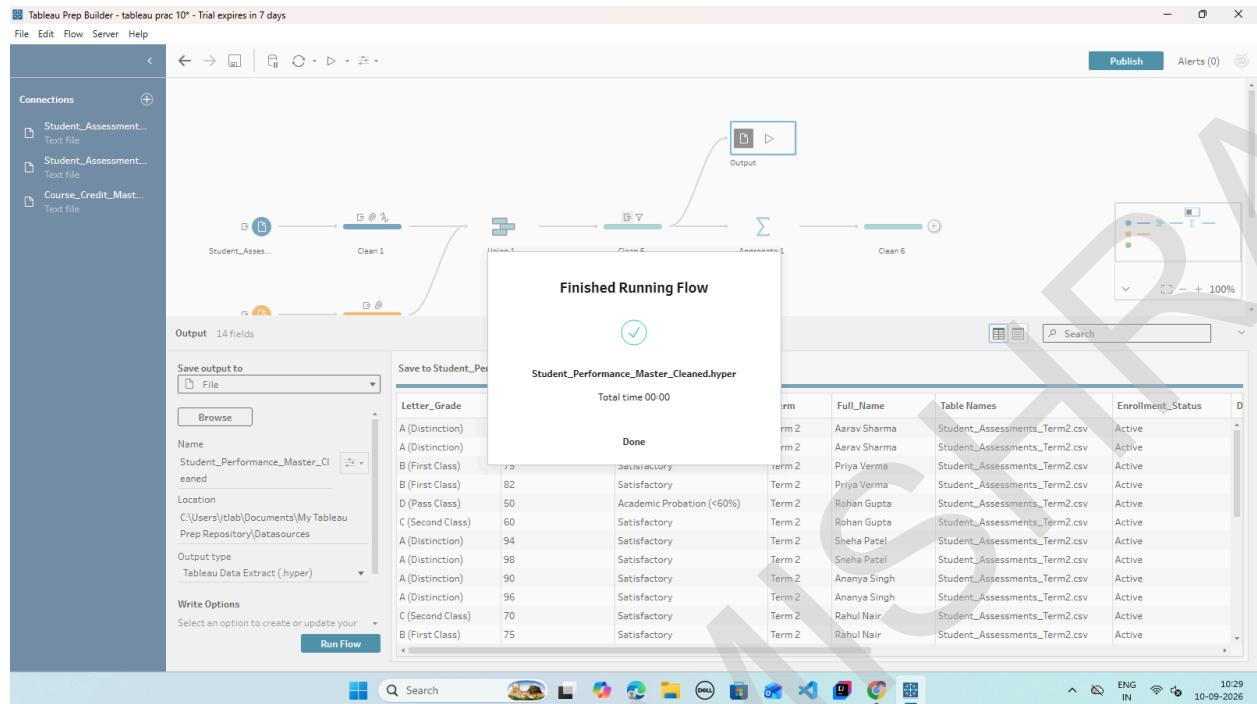


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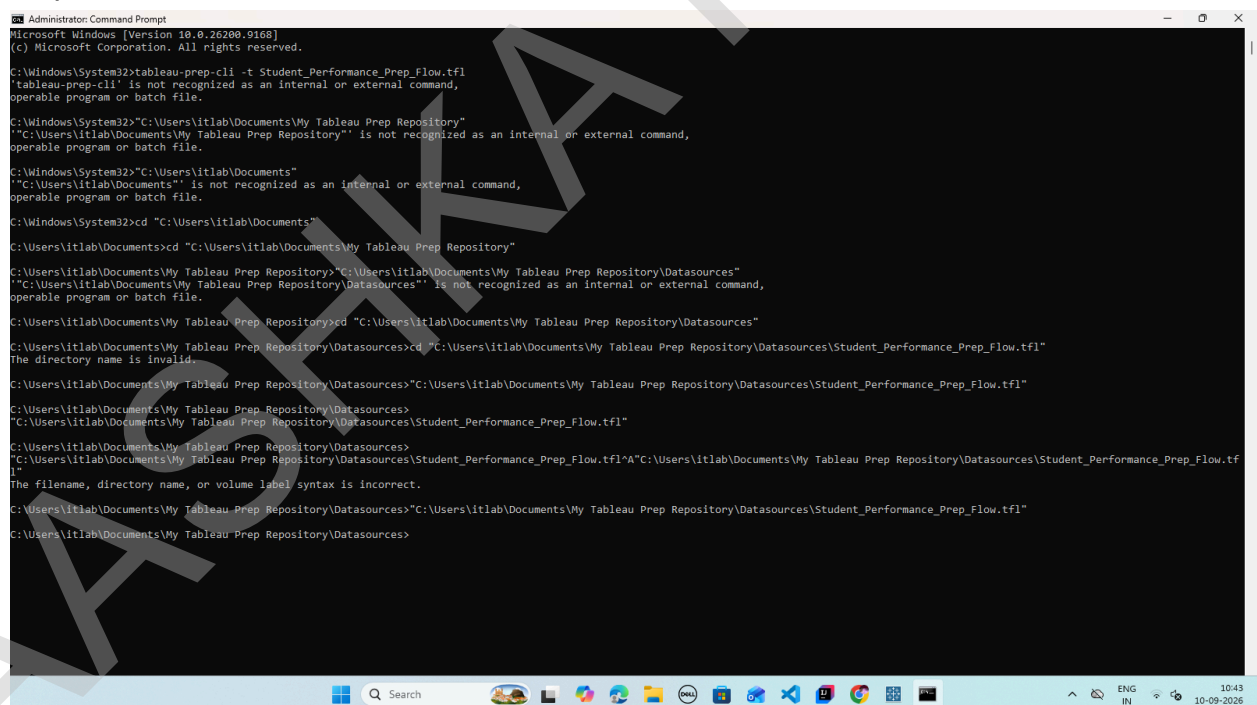
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Step 3: Run the Data Prep Flow



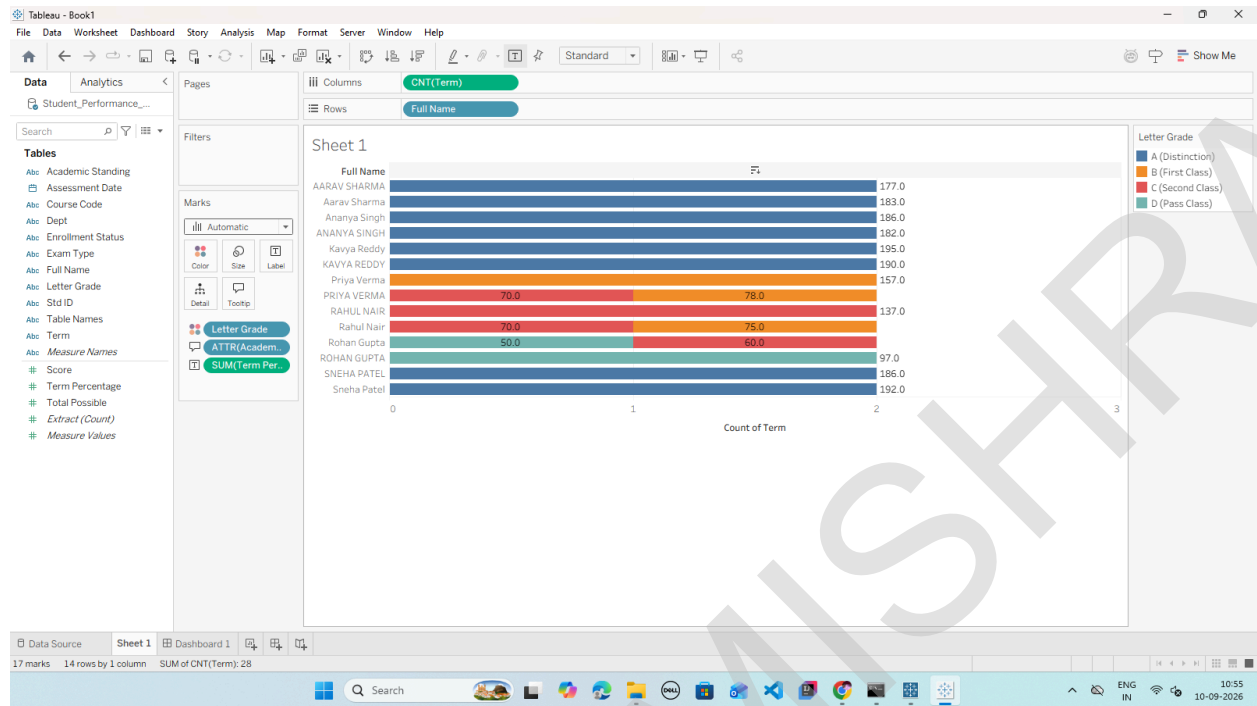
Step 4: Save and Automate the Flow File



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Step 5: Verify the Final Master Dataset in Tableau Desktop



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